

KALYANI CHARITABLE TRUST'S
LATE G. N. SAPKAL COLLEGE OF ENGINEERING,
NASHIK

DEPARTMENT OF ELECTRONICS &
TELECOMMUNICATION ENGINEERING



REPORT ON ELECTRONICS SKILL DEVELOPMENT

**A MINI PROJECT NAME: - ARDUINO BASED ULTRASONIC RADAR
SYSTEM**

NAME: - ADITYA SHARAD PAPAL

ATHRAV PRAKASH NAWALE

HITESH AVINASH THAKARE

BRANCH: - ELECTRONICS & TELECOMMUNICATION

YEAR: - SE

ROLL NUMBER: - S2 42,38,62

GUIDE BY: - PROF.S.S JAGTAP MAM

ABSTRACT

The Arduino Based Ultrasonic Radar System is an innovative and cost-effective project designed to detect objects and measure their distance using ultrasonic sensing technology. The main objective of this system is to simulate the basic working principle of a radar system by continuously scanning the surrounding environment and identifying obstacles in real time.

This system is built using an Arduino microcontroller, an ultrasonic sensor, and a servo motor. The ultrasonic sensor works by transmitting high-frequency sound waves and receiving the reflected echo signals from nearby objects. By calculating the time interval between the transmission and reception of the signal, the distance of the object is determined accurately. The servo motor rotates the ultrasonic sensor over a specific angle, typically from 0° to 180° , allowing the system to scan a wide area.

The collected data is processed by the Arduino and sent to a computer, where it is displayed in a graphical format using software such as Processing. This visual representation mimics a radar screen, showing the position and distance of detected objects in real time. The system is capable of detecting both stationary and moving objects within a limited range.

This project has several practical applications, including obstacle detection in robotics, security and surveillance systems, and automation technologies. It offers advantages such as low cost, simple design, ease of implementation, and reliable performance.

In conclusion, the Arduino Based Ultrasonic Radar System demonstrates an efficient and user-friendly approach to object detection and distance measurement, making it a valuable project for engineering students and beginners in embedded systems.

1.INTRODUCTION

In today's world, object detection and distance measurement systems play a very important role in various fields such as robotics, automation, security, and surveillance. Radar systems are widely used for detecting objects and determining their position, but traditional radar systems are complex and expensive. To overcome these limitations, a simple and cost-effective solution can be developed using ultrasonic sensor technology.

The Arduino Based Ultrasonic Radar System is a project that demonstrates the basic concept of radar using easily available electronic components. This system uses an ultrasonic sensor to detect objects and measure their distance by sending and receiving sound waves. An Arduino microcontroller is used to control the entire system, process the data, and communicate with the display interface.

In this project, a servo motor is used to rotate the ultrasonic sensor over a fixed angle, allowing the system to scan a wide area. As the sensor moves, it continuously checks for objects and calculates their distance. The collected data is then displayed on a computer screen in a graphical format, which resembles a radar screen, providing a clear visualization of the surroundings.

This project is simple, efficient, and economical, making it suitable for students and beginners to understand the fundamentals of radar systems, sensors, and embedded systems. It also has practical applications in obstacle detection, autonomous vehicles, and security systems.

Overall, the Arduino Based Ultrasonic Radar System is an excellent example of combining hardware and software to create a real-time object detection and monitoring system.

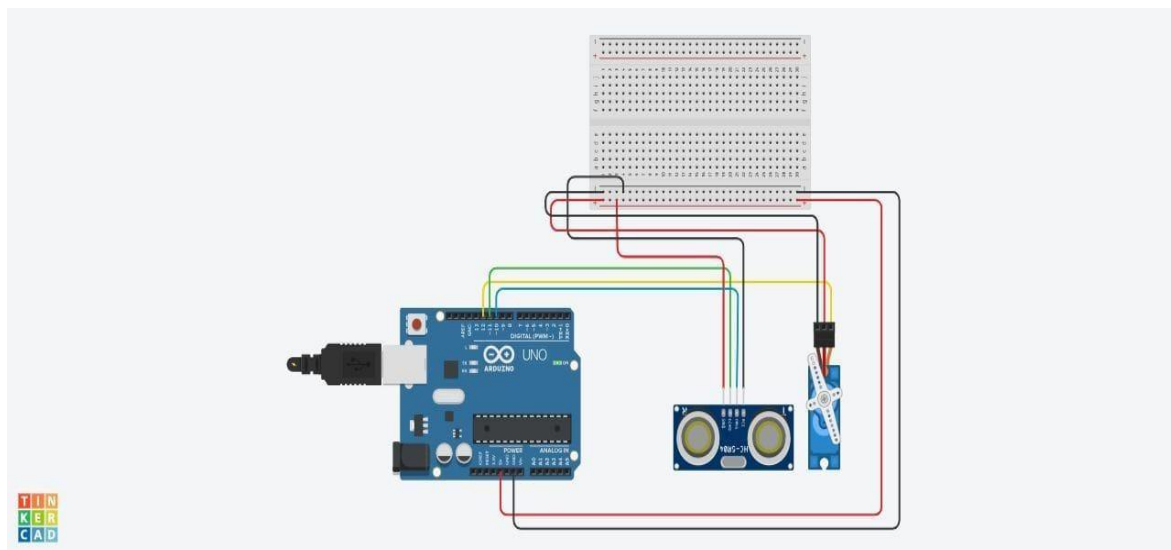
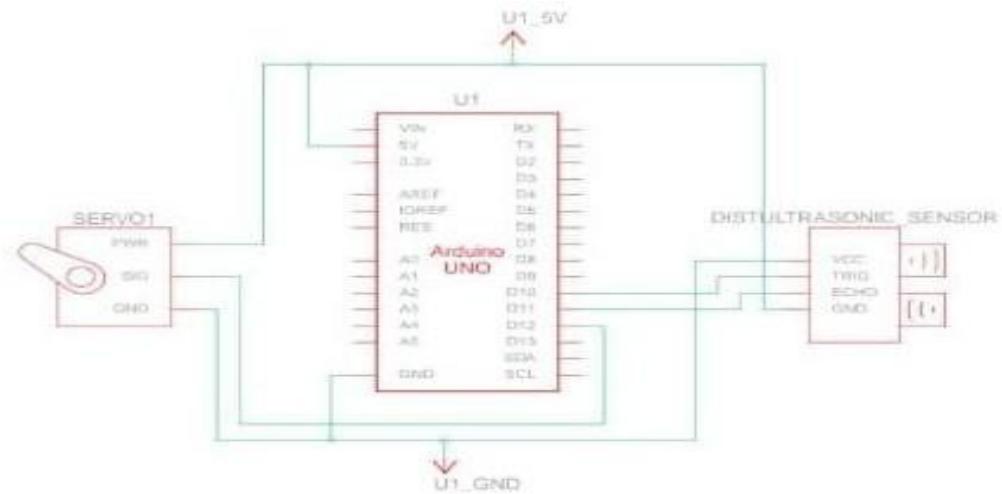
2.COMPONENTS / DEVICES REQUIRED

- **Arduino Uno:** Arduino Uno is the main controller of the system.It receives data from the sensor and controls the servo motor. It is based on ATmega328P microcontroller. Programming is done using Arduino IDE.
- **Ultrasonic Sensor (HC-SR04):** It is used to measure the distance of objects. It sends ultrasonic sound waves and receives echo. Distance is calculated using time of echo. Range is approximately 2 cm to 400 cm
- **Servo Motor (SG90):** It is used to rotate the ultrasonic sensor. It can rotate from 0° to 180°. It helps in scanning the area like radar.
- **Jumper Wires:**Used to connect different components. Available as male-to-male and male-to-female wires.
- **Breadboard:** Used for making circuit connections. No soldering is required. Helpful for testing and modifications.
- **USB Cable:** Used to provide power to Arduino. Also used for uploading the code.
- **Computer/Laptop:** Used for programming Arduino. Used to display radar output.
- **Arduino IDE Software:** Used to write and upload code. It is the main programming software.
- **Processing Software:**Used to create radar graphical display. Shows real-time object position.
- **Buzzer:** It is a sound producing device. It gives alert when object is detected. Controlled by Arduino

3. WORKING

- Arduino is the main control unit of the system.
- The system starts when power is supplied.
- Servo motor rotates the sensor from 0° to 180°.
- Ultrasonic sensor sends high-frequency sound waves.
- Waves reflect back after hitting an object.
- Sensor receives the echo signal.
- Arduino calculates distance using time.
- Distance calculation is based on speed of sound.
- Object position and distance are detected at each angle.
- System performs continuous scanning.
- Data is sent to computer using serial communication.
- Processing software displays radar output.
- Objects are shown based on angle and distance.
- Buzzer turns ON when object is near.
- Buzzer turns OFF when object is far.
- Both moving and stationary objects can be detected.
- The process repeats continuously.

4. ULTRASONIC RADAR SYSTEM BLOCK DIAGRAM



5.SIMULATION

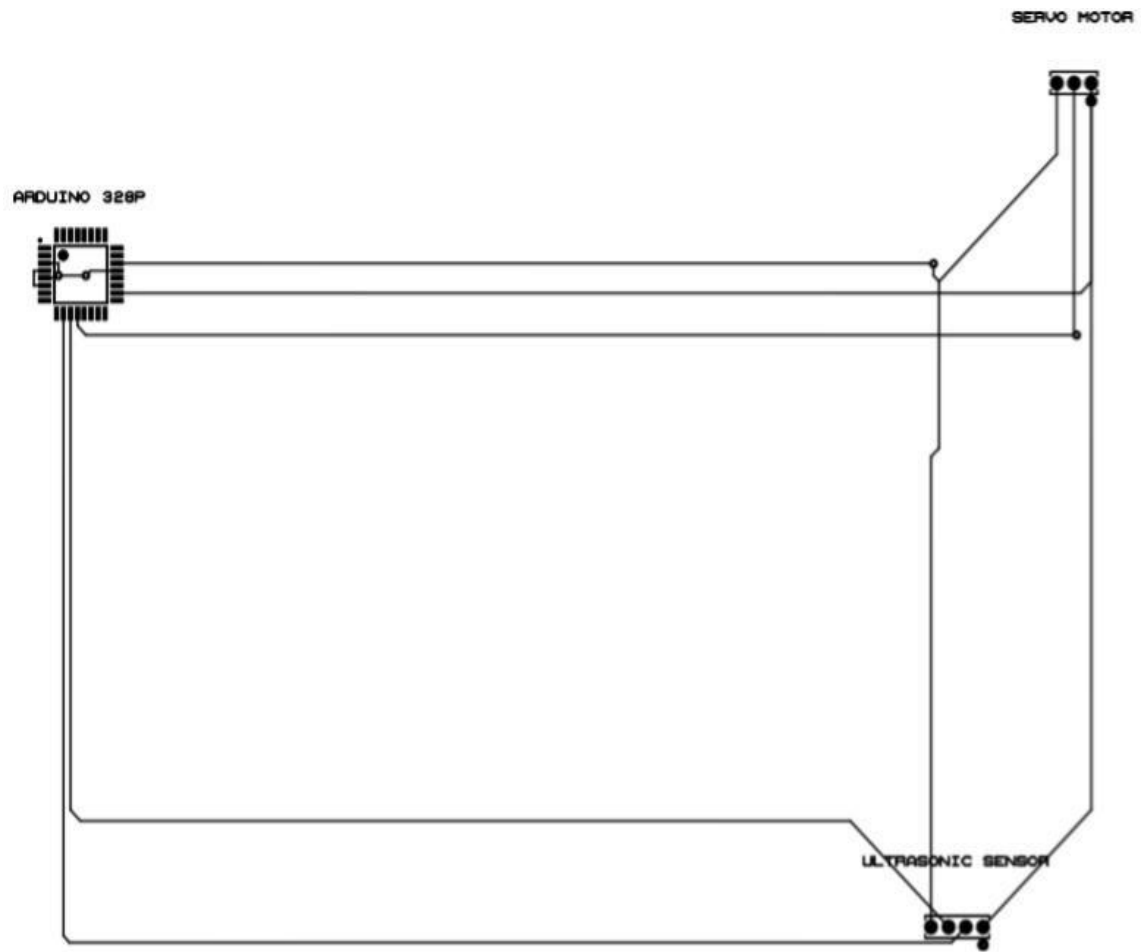
- Simulation is used before practical implementation.
- It helps to test the circuit virtually.
- Proteus or Tinkercad software can be used.
- All components are connected in simulation.
- Arduino code is uploaded in simulation.
- Working of system is verified.
- Errors and faults are identified.

6. TESTING

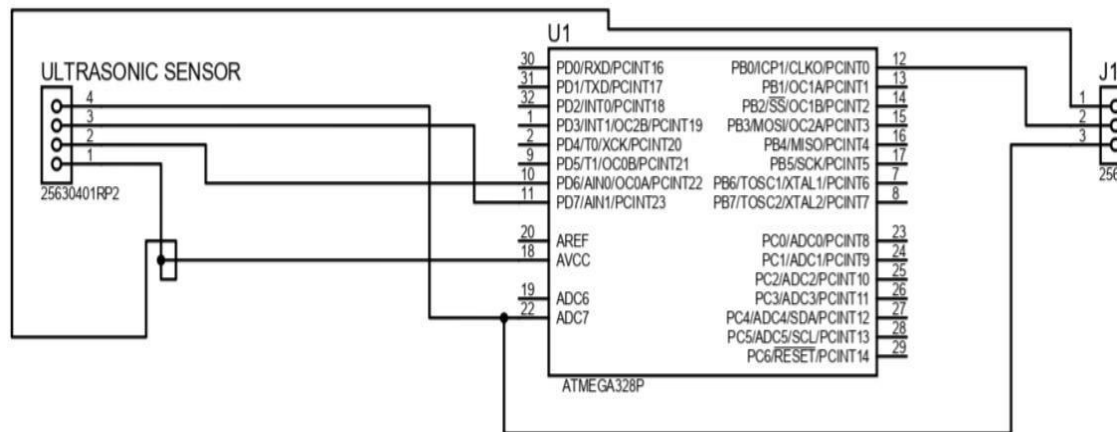
- Components are connected properly.
- Power supply is given to the system.
- Distance measurement is checked.
- Servo motor rotation is verified.
- Radar display output is observed.
- Buzzer working is tested.
- Accuracy is checked at different distances.
- Overall performance is analyzed.

7. PCB DESIGN

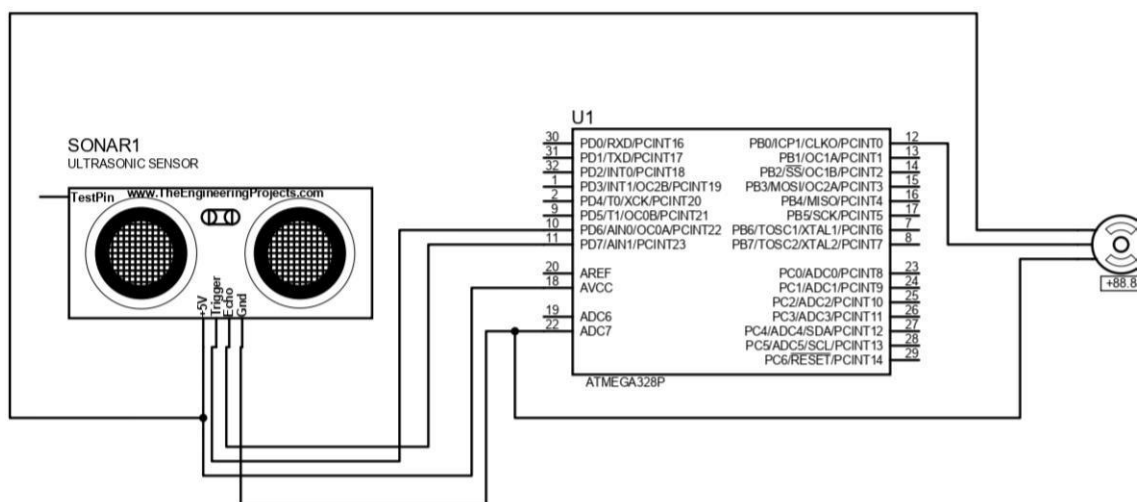
PCB DESIGN DIAGRAM



CIRCUIT DIAGRAM 1



CIRCUIT DIAGRAM 2



8.CODE

```
#include <Servo.h>

Servo radarServo;

int trigPin = 10;
int echoPin = 11;

long duration;
int distance;

void setup() {
  radarServo.attach(12);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop() {

  for(int angle = 0; angle <= 180; angle++) {
    radarServo.write(angle);
    delay(30);

    distance = getDistance();

    Serial.print(angle);
    Serial.print(",");
    Serial.print(distance);
    Serial.print(".");
  }

  for(int angle = 180; angle >= 0; angle--) {
    radarServo.write(angle);
    delay(30);

    distance = getDistance();

    Serial.print(angle);
    Serial.print(",");
    Serial.print(distance);
    Serial.print(".");
  }
}

int getDistance() {

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  return distance;
}
```

9.ADVANTAGES

- Low cost and economical system
- Simple design and easy to implement
- Real-time object detection
- Accurate distance measurement
- Works in dark environment
- Low power consumption
- Portable and compact system
- Easy maintenance
- Useful in robotics and automation
- Used in safety and security systems
- Effective for obstacle detection
- Continuous scanning capability
- Contactless sensing
- Reduces human effort
- Fast response time
- Easy to modify and upgrade
- Provides real-time monitoring
- Best for educational purposes
- Reliable and efficient system

10.APPLICATION

- Used in robotics for obstacle detection
- Used in autonomous vehicles
- Used in security systems
- Used in surveillance systems
- Used in blind assistance systems
- Used in parking systems
- Used in industrial automation
- Used in distance measurement
- Used in traffic monitoring
- Used in smart home systems

11.FUTURE SCOPE

- Can be integrated with IoT
- Can be controlled using mobile app
- Camera can be added for monitoring
- AI can be used for object detection
- Range can be increased
- Accuracy can be improved
- Wireless communication can be added
- 360° radar system can be developed
- Can be used in drones
- Can be used in smart cities

12.RESULTS

The system successfully detected objects within its specified range using the ultrasonic sensor. The distance of objects was measured accurately based on the time taken by the echo signal. The servo motor performed smooth rotation from 0° to 180°, allowing effective scanning of the surrounding area. The radar display on the computer screen clearly showed the position and distance of objects in real time.

The system achieved continuous scanning and real-time monitoring without any major delay. The buzzer functioned properly by providing an alert when objects came within a predefined distance. The system demonstrated consistent performance when tested at different distances. It was able to detect both stationary and moving objects effectively.

Overall, the system operated in a stable and reliable manner, and the output obtained was as expected according to the design of the project.

13.DATA SHEETS

Arduino Uno

The Arduino Uno is based on the ATmega328P microcontroller and operates at a voltage of 5V. It can be powered using an external supply ranging from 7V to 12V. The board consists of 14 digital input/output pins, out of which some can be used for PWM output, and 6 analog input pins. It has a clock speed of 16 MHz, which ensures efficient processing of data. The Arduino Uno is widely used due to its simplicity, reliability, and ease of programming using the Arduino IDE.

Ultrasonic Sensor (HC-SR04)

The HC-SR04 ultrasonic sensor operates at a voltage of 5V and has a working current of approximately 15 mA. It uses ultrasonic sound waves with a frequency of 40 kHz to measure distance. The sensor is capable of measuring distances in the range of 2 cm to 400 cm with an accuracy of about ± 3 mm. It works by transmitting sound waves and calculating the time taken for the echo to return after reflecting from an object.

Servo Motor (SG90)

The SG90 servo motor operates within a voltage range of 4.8V to 6V and is capable of rotating between 0° and 180° . It has a speed of approximately 0.1 seconds for every 60° rotation, making it suitable for quick positioning tasks. The motor provides a torque of about 1.8 kg/cm and is lightweight, weighing around 9 grams. It is commonly used in projects that require precise angular movement.

Buzzer

The buzzer used in this project typically operates within a voltage range of 3V to 5V and consumes very low current. An active buzzer is generally preferred as it can produce sound directly when power is supplied. It generates a simple beep sound, which is used as an alert signal in the system. The buzzer is controlled by the Arduino and is activated when an object is detected within a certain distance.

14.REFERENCES

- Arduino Official Website. Arduino Uno Documentation and Technical Specifications. Available at: <https://www.arduino.cc>
— Arduino Team
- HC-SR04 Ultrasonic Sensor Datasheet. Distance Measurement Sensor Specifications and Working Principle.
— ELECFreaks / Generic Manufacturer Datasheet
- SG90 Servo Motor Datasheet. Micro Servo Motor Technical Specifications.
— TowerPro Manufacturer
- Arduino IDE Documentation. Software Guide for Writing, Compiling, and Uploading Code.
— Arduino Team
- Processing Software Official Website. Graphical Visualization and Radar Display Development.
— Processing Foundation
- Electronics Tutorials and Online Learning Platforms. Ultrasonic Sensor and Embedded Systems Concepts.
— Various Authors / Online Contributors
- Basic Concepts of Radar Systems. Engineering Study Material and Notes.
— Various Authors
- Educational YouTube Channels and Online Videos. Arduino Radar Project Implementation.
— Various Content Creators